AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102
Physics

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 18 /Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5hrs

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple Choice Questions (MCQ). [15 Marks]

1. The number 3.00 × 10^5 has how many significant figures?
A. 1
B. 2
C. 3
D. 5

2. The prefix "micro-" represents:
A. 10^{-3}
B. 10^{-6}
C. 10^{-9}
D. 10^{6}

3. The dimensional formula of force is:
A. [M L T^{-2}]
B. [L T^{-2}]
C. [M L^{2} T^{-1}]
D. [M L T^{-1}]

4. Dimensional analysis is primarily used to:
A. Find numerical values
B. Check equation consistency
C. Measure precision
D. Increase accuracy

5. When solving physics problems, identifying known and unknown quantities is part of:
A. Guessing
B. Approximation
C. Problem-solving technique
D. Dimensional checking

6. In approximation, 3.1416 can be rounded to 3.14 because:
A. It increases precision
B. It simplifies calculations with minimal error
C. It changes the result completely
D. It makes the number larger

7. A force that always opposes motion is:
A. Normal force
B. Gravitational force
C. Friction
D. Tension

8. The graphical method of vector addition is based on:
A. Triangle and polygon laws
B. Differentiation
C. Integration
D. Dimensional analysis

9. The component of a 1st quarter vector with angle θ with x-axis, along the x-axis is found using:
A. sinθ
B. cosθ
C. tanθ
D. cotθ

10. Newton's First Law implies that:
A. An unbalanced force causes rest
B. No net force means no acceleration
C. Force is proportional to acceleration
D. Force changes inertia

11. The displacement of an object is zero if it:
A. Travels at constant speed
B. Returns to the starting point
C. Moves in a circle
D. Accelerates uniformly

12. Velocity is a:
A. Scalar
B. Vector
C. Constant
D. Unitless quantity

13. The SI unit of acceleration is:
A. m/s
B. m/s²
C. N
D. kg·m/s

14. Newton's Second Law relates force to:
A. Momentum
B. Displacement
C. Acceleration
D. Energy

15. The net force on a 5 kg object accelerating at 2 m/s² is:
A. 2.5 N
B. 7 N
C. 10 N
D. 15 N

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16. If a car accelerates from rest at 4 m/s² for 5 s, its final velocity is:
A. 5 m/s
B. 10 m/s
C. 15 m/s
D. 20 m/s

17. The equation $\Delta s = ut + \{1/2\} a t^2$ gives:
A. Velocity
B. Displacement
C. Momentum
D. Force

18. A stone dropped from a height falls under:
A. Constant acceleration
B. Constant velocity
C. Increasing force
D. Decreasing momentum

19. In free fall, neglecting air resistance, heavier and lighter objects:
A. Fall at different rates
B. Fall at the same rate
C. Stop midway
D. Accelerate differently

20. Momentum has both magnitude and:
A. Energy
B. Direction
C. Speed
D. Mass

21. The total momentum before and after a collision remains the same if:
A. There is no external force
B. Kinetic energy is not conserved
C. Mass changes
D. Gravity is acting

22. The impulse of a 10 N force acting for 3 s is:
A. 3 N·s
B. 13 N·s
C. 30 N·s
D. 33 N s

23. The unit N·s is equivalent to:
A. kg·m/s
B. J/s
C. W
D. m/s²

24. In a head-on elastic collision, the relative velocity of approach equals:
A. Zero
B. Relative velocity of separation
C. Sum of individual speeds
D. Difference in kinetic energy

25. A completely inelastic collision is one where:
A. Kinetic energy is conserved
B. The objects stick together
C. No momentum is exchanged
D. The objects bounce perfectly

26. Which law states that an object's momentum remains constant if no net force acts on it?
A. Law of Inertia
B. Conservation of Momentum
C. Impulse Theorem
D. Hooke's Law

27. A projectile's horizontal velocity remains constant when:
A. Air resistance acts
B. Gravity is ignored
C. No net horizontal force acts
D. Mass changes

28. An object thrown vertically upward has zero velocity:
A. At the start
B. At the highest point
C. Throughout its motion
D. Just before hitting the ground

29. When two equal and opposite forces act on an object, the body is said to be in:
A. Motion
B. Equilibrium
C. Rotation
D. Free fall

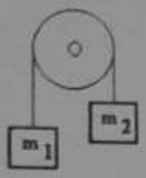
30. The term "vector addition using components" means:
A. Adding magnitudes only
B. Adding corresponding x and y components
C. Multiplying vectors together
D. Subtracting scalar quantities

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Part II: Answer the following questions. [15 Marks]

Question 1: (5 Marks)
Two masses are being pulled up a 30.0 degree incline by a force F parallel to the incline. The acceleration up the incline is 1.00 m/s^2 and the velocity is up the incline. The force is applied to a 200 kg mass, and a string connects the 200 kg mass to a 150 kg mass. The coefficient of kinetic friction is 0.200. The tension in the string is?

Question2 (5 marks)
Two masses are suspended by a cord that passes over a pulley with negligible mass. The cord also has negligible mass. One of the masses, m_1, has a mass of 5.0 kg and the other mass, m_2, has a mass of 3.0 kg. The acceleration of m_1 is



Question3 (5marks)
Two small steel balls A and B have mass 0.6 kg and 0.2 kg respectively. They are moving towards each other in opposite directions on a smooth horizontal table when they collide directly. Immediately before the collision, the speed of A is 8 m/s and the speed of B is 2 m/s. Immediately after the collision, the direction of motion of A is unchanged and the speed of B is twice the speed of A. Find;
(a) the speed of A immediately after the collision,
(b) the magnitude of the impulse exerted on B in the collision.

End of questions

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